



Instructor: Date:

Learner: Regional Branch:

Program: Center:

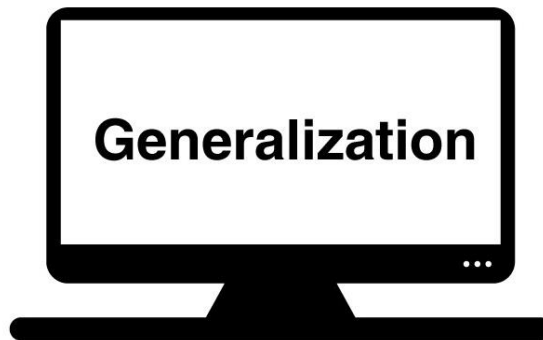
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Let's Start!

A Crash Course on ER Model



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to relational database management system in spoken or written texts.

Describe relational database, its concepts, and principles in spoken or written texts.

Can Do Statements:

- I can show understanding of the concepts of generalization and specialization in the ER model.
- I can depict information in diagrams about generalization and specialization in the ER model.
- I can discuss the concept of inheritance in the ER model.

Ask your partner:

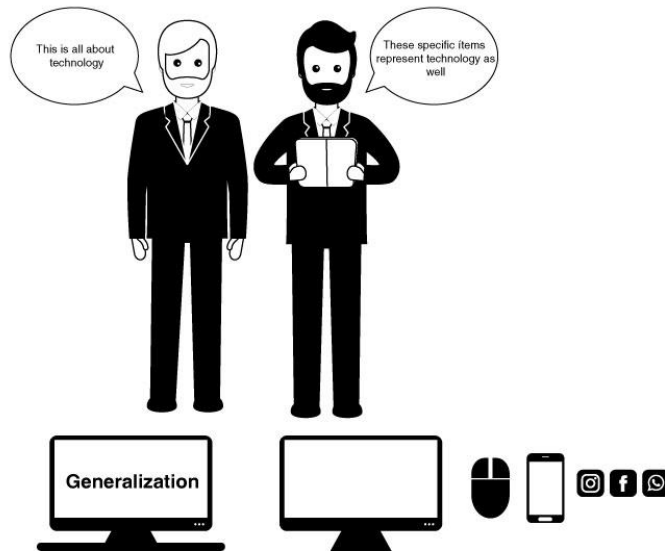
- What do you know about the ER model?
- How useful do you think a hierarchical classification model is?





Let's Explore!

A Conversation between Co-workers



Instructions: Read the text below.

Greg: What's up Matt? What are you doing?

Matt: Umm I'm working on the ER model for a school database, but I'm stuck in the inheritance part, it's kind of tricky!

Greg: Are the concepts of generalization and specialization confusing for you too?

Matt: Well, I understand both. In the ER database model there is a hierarchical manner to organize the entities; check the information in this diagram. As you can see, it can be done in two ways, generalization or specialization. Generalization is where an entity can be connected or can group a number of entities that are similar. The specialization is very different; it means that a group of entities is distributed into sub-groups taking into account their characteristics.

Greg: You have a good understanding on that part, but can some data be generalized, and at the same time, be specialized?

Matt: I don't know; I don't think so. Can this job be done in spite of ignoring that?

Greg: Not really! That is exactly what Inheritance is! In the hierarchy, as you go up, you see generalization and as you go down you see the detail of every entity, all these details are not on sight for the users, this is the abstraction. Then, as everything is connected, the Inheritance allows the low-level entities to receive the attributes of the high-level ones.

Matt: I get it now! For example, if we are talking about a school, the characteristics of a person like a name or address can be received by lower level entities like staff or administrative.

Greg: You got it!



**Let's Practice!**

Important concepts ought to be clarified right away.

Instructions: Classify the sentences by writing the number of the sentence in the correct category. Check your answers with your partner. Discuss.

Generalization and Specialization concepts can be better defined based on the characteristics and functions of each one. Though, some of them can be shared.

1. ER database model organization
2. Part of the entities' hierarchical organization.
3. It groups entities
4. Similar characteristics
5. It sub-groups entities
6. Entities distributed

Generalization	Specialization

**Let's Practice!**

The ER model classifies the entities from general to specific and the other way around.

Instructions: Match column A with column B. Write sentences with column A concepts.

COLUMN A

1. Icon
2. Email icon
3. Development
4. Web development
5. Diagrams
6. Entity Relationship Model diagram

COLUMN B

- a. Specialization
- b. Generalization
- c. Specialization
- d. Generalization
- e. Generalization
- f. Specialization



Extension Activity

I can understand more about inheritance and generalization.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about inheritance and generalization.
- List the main attributes of inheritance and generalization.
- Create a poster with the main characteristics of inheritance and generalization.
- Share with your partners the post you created.

Self-Assessment

Yes

No

- I look for more information about inheritance and generalization. ☐ Yes ☐ No
- I list the main attributes of inheritance and generalization. ☐ Yes ☐ No
- I create a post with the main characteristics of inheritance and generalization. ☐ Yes ☐ No
- I share with your partners the post I created. ☐ Yes ☐ No



Target Words

- **manner** (noun): the way or method in which something is done.
e.g. There is a hierarchical manner to organize the entities.
- **generalize** (verb): to make an overall idea about something
e.g. You have a good understanding on that part, but can some data be generalized, and at the same time, be specialized?
- **group** (verb): to form a group by integrating people or things all together
e.g. Generalization is where an entity can be connected or can group a number of entities that share some similar characteristics.
- **distribute** (verb): the way to arrange things in a specific order.
e.g. it means that a group of entities is distributed into sub groups.
- **specialize** (verb): that follows a special line of study or order.
e.g. Can some data be generalized, and at the same time, be specialized?
- **on sight** (expression): that can be seen easily.
e.g. All these details are not on sight for the users this is the abstraction.



Learning Points

Passive Modals Interrogative

The **passive voice** is used to give emphasis to the object instead of the subject.

The **passive voice with modals** to make questions is used to know information about the object.

The form is like this:
(chart)

Modal verb	object	be	Verb in past participle?
<i>Could</i>	<i>the project</i>	<i>be</i>	<i>designed without taking into account generalization and specialization?</i>
<i>Can</i>	<i>generalization</i>	<i>be</i>	<i>made about people?</i>
<i>Must</i>	<i>specialization</i>	<i>be</i>	<i>understood as the structure that goes down?</i>

Examples

- *Can some data be generalized, and at the same time, be specialized?*
- *Can this job be done in spite of ignoring that?*

Distinguishing generalization from specialization

The **ER model** can define the entities in a structural way coming from the general to the specific and the other way around.

There are two structures used on the ER model the generalization and the specialization. Both of these structures are different as the generalization a number of entities respond to a generalize entity that is based on their similar characteristics, like if I say poodle, German shepherd and cocker spaniel they can all be generalized as dogs.

The other structure is the specialization. Which is contrary to generalization, in this case a group of entities is separated into other small groups depending on their characteristics. For example, in a school a person can be a student, a teacher, an administrative or staff.

Let's see the graphs:

